Terminal-Bench: What Researchers Look For in a Dataset

Simple metric / quality guide for task acceptance
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Purpose

This short guide focuses specifically on the question: **“What metrics and quality characteristics do researchers look for when deciding whether a Terminal-Bench task is suitable for the dataset?”**

Important: Terminal-Bench's public TB3 task-implementation rubric contains 28 named criteria. The items below highlight the most important researcher-facing dimensions for understanding dataset/task quality; they are not a claim that these are the only 28 checks.

1. The Main Metrics / Criteria

Metric / Criterion	What researchers want	Simple example
Verifiability	Can we reliably determine if the agent completed the task correctly?	Example: tests check whether the requested file contains the correct output.
Solvability	Can the task actually be completed by a valid solution?	Example: an expert runs the reference solution and all required tests pass.
Difficulty	Is the task genuinely difficult for an AI agent and requires meaningful expertise?	Example: debugging a multi-component system is stronger than answering a simple syntax question.
Agentic / Multi-step Work	Does the task require the agent to inspect, reason, act, and verify through the terminal?	Example: inspect logs → identify cause → edit configuration → run tests → verify result.
Outcome Verification	Does verification judge the final result rather than forcing one particular process?	Example: any valid method is accepted if the final application state is correct.
Functional Verification	Do tests execute/check the actual functionality?	Example: run the program and verify its behavior instead of only searching for a keyword in the source.
Anti-cheat Robustness	Can the agent pass without genuinely solving the task?	Example: tests should resist hardcoded answers, modifying tests, fake wrappers, or answer leakage.
Determinism / Reproducibility	Will the task behave consistently across runs?	Example: pin important dependencies/data instead of relying on changing live services.
Instruction–Test Alignment	Do the tests verify exactly what the instructions require?	Example: if the instruction requires three outputs, the verifier should check those required outputs.
Novelty	Is the task more than a routine/textbook or memorization exercise?	Example: a realistic engineering workflow is stronger than a copied beginner exercise.
Security	Is the task safe and free from malicious or unsafe mechanisms?	Example: no credential exfiltration, host escape, destructive operations, or prompt injection.
Resource Suitability	Are timeout, CPU, memory, storage and GPU resources appropriate?	Example: the task should be hard because of reasoning/work, not because the container has too little memory.

2. The Simple Idea Behind These Metrics

Researchers are essentially asking four questions:

1. Is it a good task? — realistic, useful, novel and genuinely difficult.

2. Can an AI agent actually work on it? — it requires meaningful multi-step terminal interaction.

3. Can we fairly judge the result? — reliable, functional, deterministic verification with good instruction–test alignment.

4. Can the benchmark be trusted? — secure, resistant to cheating, reproducible and appropriately resourced.

3. One Complete Example

Example task: “Fix a broken web application inside the provided container and make all tests pass.”

Criterion	How the example satisfies it
Difficulty	Agent must inspect logs, understand the failure and choose the correct fix.
Agentic	Multiple terminal steps: inspect → diagnose → edit → run → debug → verify.
Verifiable	Automated tests determine whether the application works.
Functional verification	Tests actually run the application and check its behavior.
Outcome verification	The agent can use any valid fix; only the correct final state matters.
Anti-cheat	The agent cannot simply edit tests or hardcode the expected result.
Reproducible	The environment and important dependencies are controlled/pinned.
Instruction-test alignment	Every required behavior in the instruction has corresponding verification.
Security	The task does not require or introduce credential theft, host escape, or other malicious behavior.
Resources	The container has enough time/memory/CPU for the task without making the challenge artificially difficult.

4. What the Researcher Ultimately Wants

A task that is hard for the AI, easy enough for the computer to run, and reliable enough for researchers to measure.

In other words: **realistic task + genuine difficulty + agentic work + reliable verification + security + reproducibility + anti-cheating.**

5. Important Benchmark Metric

Resolution Rate is the core benchmark outcome: the percentage of evaluated tasks that an agent successfully resolves under the benchmark's verification.

For example, if an agent successfully resolves 40 out of 100 evaluated tasks, its resolution rate is 40%. When comparing scores, always consider the benchmark version, task set and evaluation setup.

6. Short Answer for Your Team Lead

“Sir, the main things researchers look for are whether the task is genuinely difficult and realistic, requires multi-step agentic work, can be reliably and functionally verified, is reproducible, aligned between instructions and tests, resistant to cheating, secure, novel, and given appropriate resources. The overall benchmark outcome is resolution rate.”

Sources

Terminal-Bench public TB3 task-implementation rubric: https://raw.githubusercontent.com/scaleapi/terminal-bench-3-public/main/rubrics/task-implementation.toml

Terminal-Bench TB3 contribution call: https://www.tbench.ai/news/tb3-contribution-call

Terminal-Bench benchmark overview: https://www.tbench.ai/news/announcement

Terminal-Bench leaderboard: https://www.tbench.ai/leaderboard